

Marketing Growth Analytics

OCTOBER 2026

Presented by Muhammad Alfi Syahputra

Introduction



Summary

Graduated from President University majoring in International Relations with a broad understanding of global affairs, Indonesian politics, gender issues, and popular culture. A self-motivated self-starter that has improved his expertise and grown professionally. Highly interested in Data Analyst, with skills in data processing, building data pipelines, and big data analysis. Proficient in Python, SQL, and tools such as Tableau and PowerBI. Known for being detail-oriented, adaptable, and capable of working both independently and collaboratively. Ready to leverage prior work experience to support data management and optimization, contributing to data-driven decision-making.

Education

Dibimbing Boothcamp as Data Analyst **2026 - Present**

- Studied data storytelling, visualization, and performance measurement using industry tools.
- Technical Skills: Excel (pivot tables, data cleaning), SQL, Tableau, Power BI, Google Collab.
- Capstone Project: Designed an interactive dashboard to analyze company performance metrics and KPIs.

President University /International Relations / GPA (3.49 / 4.00) **2018 - 2022**

Skills

- **Data Analyst**

Hard Skills:

Data Cleaning & Processing: Excel (advanced), SQL queries, data validation

Business Reporting: dashboard creation, KPItracking, ad-hoc reports

Data Visualization: Power BI, Google Data Studio, Tableau (basic)

Quantitative Analysis: identifying trends and correlations for business decisions

Toolset: Excel, Power BI, SQL, Google Sheets, Google Data Studio

- **Languages: English (Professional proficiency), Japanese (Beginner) N5**
- Achievements (2018): Outstanding Performance as the participant in Diplomat One Day
- Achievements (2018): The Best Participant in Table Manner out of 150 participants in Diplomat One Day
- **Skills : Public Speaking, Content Writing, Event Communication, Design & Editing Photo, Editing Video**

(Microsoft Office, Canva, Google Docs, Slides, and Presentation, Filmora)

Project

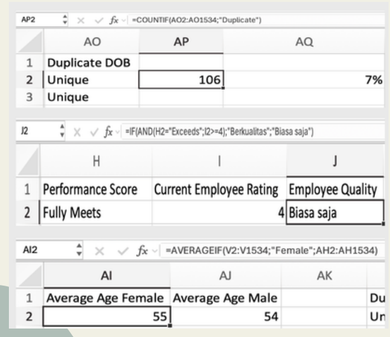


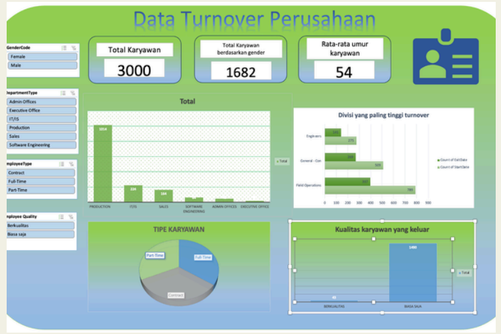


Social Project (Education Support)



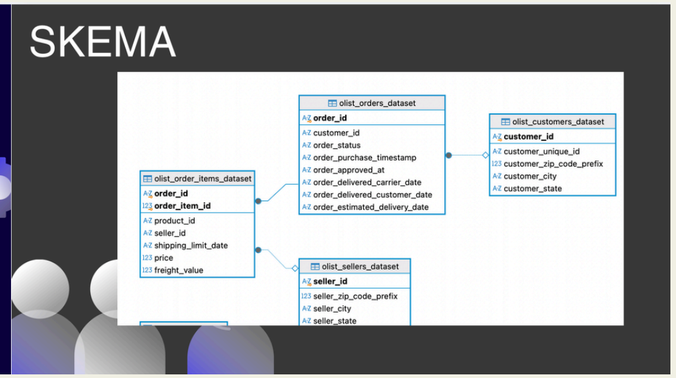
Bigmart sales Analysis (Python)





Fenomena Turnover (Excel)





Olist E-Commerce Analysis (SQL)

Journal

The Implementation of Data Analytics in Digital Marketing to Improve Business Performance

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The integration of data analytics into digital marketing strategies can significantly improve business performance. By leveraging consumer insights, optimizing campaigns, and personalizing interactions, companies can achieve better engagement, higher ROI, and stronger brand loyalty. However, the journey toward data-driven marketing success requires overcoming technical, organizational, and ethical hurdles. A strategic commitment to analytics, backed by proper infrastructure and skills development, is essential for sustainable competitive advantage.

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Marketing Strategy

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Problem Statement



- Channel marketing mana yang paling efektif menghasilkan pembelian?
- Di tahap mana company kehilangan calon pelanggan paling banyak?
- Bagaimana tren pembelian pelanggan?

Data Understanding

	user_id	first_visit_date	channel	source	device	sessions	page_views	bounce	add_to_cart	purchase	revenue	ad_cost	clicks	impressions	repeat_purchase	retention_month
0	1	2025-02-17	Referral	instagram	Mobile	7	4	0	1	1	401318	0	40	880	0	2
1	2	2025-01-05	Referral	facebook	Mobile	3	5	0	0	0	0	0	12	264	0	0
2	3	2025-01-26	Referral	tiktok	Mobile	4	5	0	0	0	0	0	3	72	0	0
3	4	2025-02-24	Paid Ads	facebook	Mobile	4	2	0	1	0	0	18215	16	320	0	0
4	5	2025-01-04	Referral	instagram	Mobile	1	18	0	0	0	0	0	44	792	0	0

Dataset memiliki dimensi :

Jumlah Baris = 800

Jumlah Kolom = 16

```
data = pd.read_csv("marketing_growth_analytics.csv")
```

Sumber data:

https://drive.google.com/file/d/16YPDZhN2sMJHJ5zdzvWKRuyknJokaRZY/view

Column Description

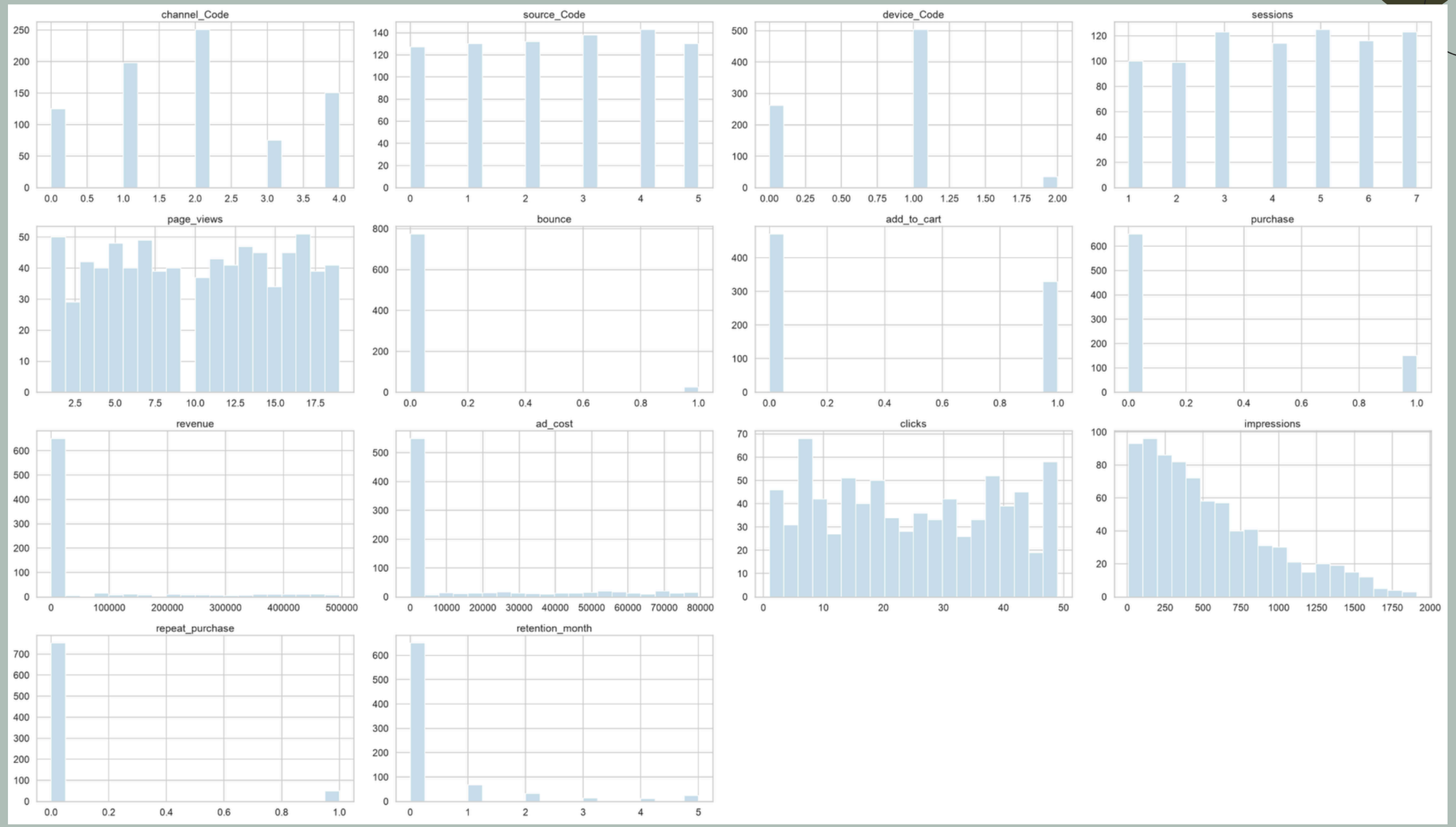
No.	Nama Kolom	Deskripsi
1	`user_id`	Identitas unik setiap pengguna dalam dataset.
2	`first_visit_date`	Tanggal pertama kali pengguna mengunjungi platform.
3	`channel`	Saluran pemasaran yang digunakan untuk memperoleh pengguna.
4	`source`	Sumber traffic yang membawa pengguna ke platform.
5	`device`	Perangkat yang digunakan pengguna untuk mengakses platform.
6	`sessions`	Jumlah sesi kunjungan yang dilakukan pengguna.
7	`page_views`	Jumlah halaman yang dilihat pengguna selama aktivitas kunjungan.
8	`bounce`	Indikator apakah pengguna mengalami bounce atau meninggalkan kunjungan tanpa interaksi lanjutan sesuai definisi dataset.
9	`add_to_cart`	Indikator apakah pengguna menambahkan produk ke keranjang belanja.
10	`purchase`	Indikator apakah pengguna melakukan pembelian.
11	`revenue`	Pendapatan yang dihasilkan dari aktivitas pengguna.
12	`ad_cost`	Biaya iklan yang terkait dengan aktivitas pemasaran.
13	`clicks`	Jumlah klik yang diperoleh dari aktivitas pemasaran atau iklan.
14	`impressions`	Jumlah tayangan iklan atau konten kepada pengguna.
15	`repeat_purchase`	Indikator apakah pengguna melakukan pembelian berulang.
16	`retention_month`	Informasi mengenai retensi pengguna dalam satuan bulan.

Data Type

```
marketing.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 16 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   user_id            800 non-null    int64
 1   first_visit_date   800 non-null    str
 2   channel            800 non-null    str
 3   source             800 non-null    str
 4   device             800 non-null    str
 5   sessions           800 non-null    int64
 6   page_views         800 non-null    int64
 7   bounce             800 non-null    int64
 8   add_to_cart        800 non-null    int64
 9   purchase           800 non-null    int64
 10  revenue            800 non-null    int64
 11  ad_cost            800 non-null    int64
 12  clicks             800 non-null    int64
 13  impressions        800 non-null    int64
 14  repeat_purchase    800 non-null    int64
 15  retention_month    800 non-null    int64
dtypes: int64(12), str(4)
memory usage: 125.8 KB
```

Data Distribution



Outlier Checking

Sessions

```
Q1 = marketing["sessions"].quantile(0.25)
Q3 = marketing["sessions"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

value_outlier = marketing[
    (marketing["sessions"] < lower) |
    (marketing["sessions"] > upper)
]

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower Bound:", lower)
print("Upper Bound:", upper)
print("Jumlah Outlier:",value_outlier["sessions"].count())
```

```
Q1: 3.0
Q3: 6.0
IQR: 3.0
Lower Bound: -1.5
Upper Bound: 10.5
Jumlah Outlier: 0
```

Page Views

```
Q1: 5.0
Q3: 15.0
IQR: 10.0
Lower Bound: -10.0
Upper Bound: 30.0
Jumlah Outlier: 0
```

Bounce

```
Q1: 0.0
Q3: 0.0
IQR: 0.0
Lower Bound: 0.0
Upper Bound: 0.0
Jumlah Outlier: 26
```

Add_to_cart

```
Q1: 0.0
Q3: 1.0
IQR: 1.0
Lower Bound: -1.5
Upper Bound: 2.5
Jumlah Outlier: 0
```

Purchase

```
Q1: 0.0
Q3: 0.0
IQR: 0.0
Lower Bound: 0.0
Upper Bound: 0.0
Jumlah Outlier: 150
```

Revenue

```
Q1: 0.0
Q3: 0.0
IQR: 0.0
Lower Bound: 0.0
Upper Bound: 0.0
Jumlah Outlier: 150
```

Ad_cost

```
Q1: 0.0
Q3: 21996.75
IQR: 21996.75
Lower Bound: -32995.125
Upper Bound: 54991.875
Jumlah Outlier: 89
```

Clicks

```
Q1: 11.0
Q3: 37.0
IQR: 26.0
Lower Bound: -28.0
Upper Bound: 76.0
Jumlah Outlier: 0
```

Impressions

```
Q1: 208.75
Q3: 812.0
IQR: 603.25
Lower Bound: -696.125
Upper Bound: 1716.875
Jumlah Outlier: 7
```

Repeat_Purchase

```
Q1: 0.0
Q3: 0.0
IQR: 0.0
Lower Bound: 0.0
Upper Bound: 0.0
Jumlah Outlier: 48
```

Retention_month

```
Q1: 0.0
Q3: 0.0
IQR: 0.0
Lower Bound: 0.0
Upper Bound: 0.0
Jumlah Outlier: 150
```

Duplicate & Missing value

```
duplicate_persentase = marketing.duplicated().mean() * 100

print(f"Persentase duplicate: {duplicate_persentase:.2f}%")
```

```
Persentase duplicate: 0.00%
```

```
marketing_clean = pd.DataFrame({
    "data_missing": marketing.isna().sum(),
    "persentase": (marketing.isna().mean()*100).round(2)
})

marketing_clean = marketing_clean[marketing_clean["data_missing"]>0]
marketing_clean
```

	data_missing	persentase

Tidak terdapat missing value pada data tersebut

Early Insight

Insight dari data exploration yang bisa disimpulkan yaitu, dataset memiliki variabel numerik dan kategorikal, tidak ditemukannya missing value, pada kolom first_visit_date perlu dibuah tipe datanya supaya menjadi format datetime agar lebih sesuai dan bukan string. Terdapat outlier namun tidak perlu ditangani karena hal tersebut merupakan hal yang wajar dalam kolom-kolom tersebut

Data Preparation

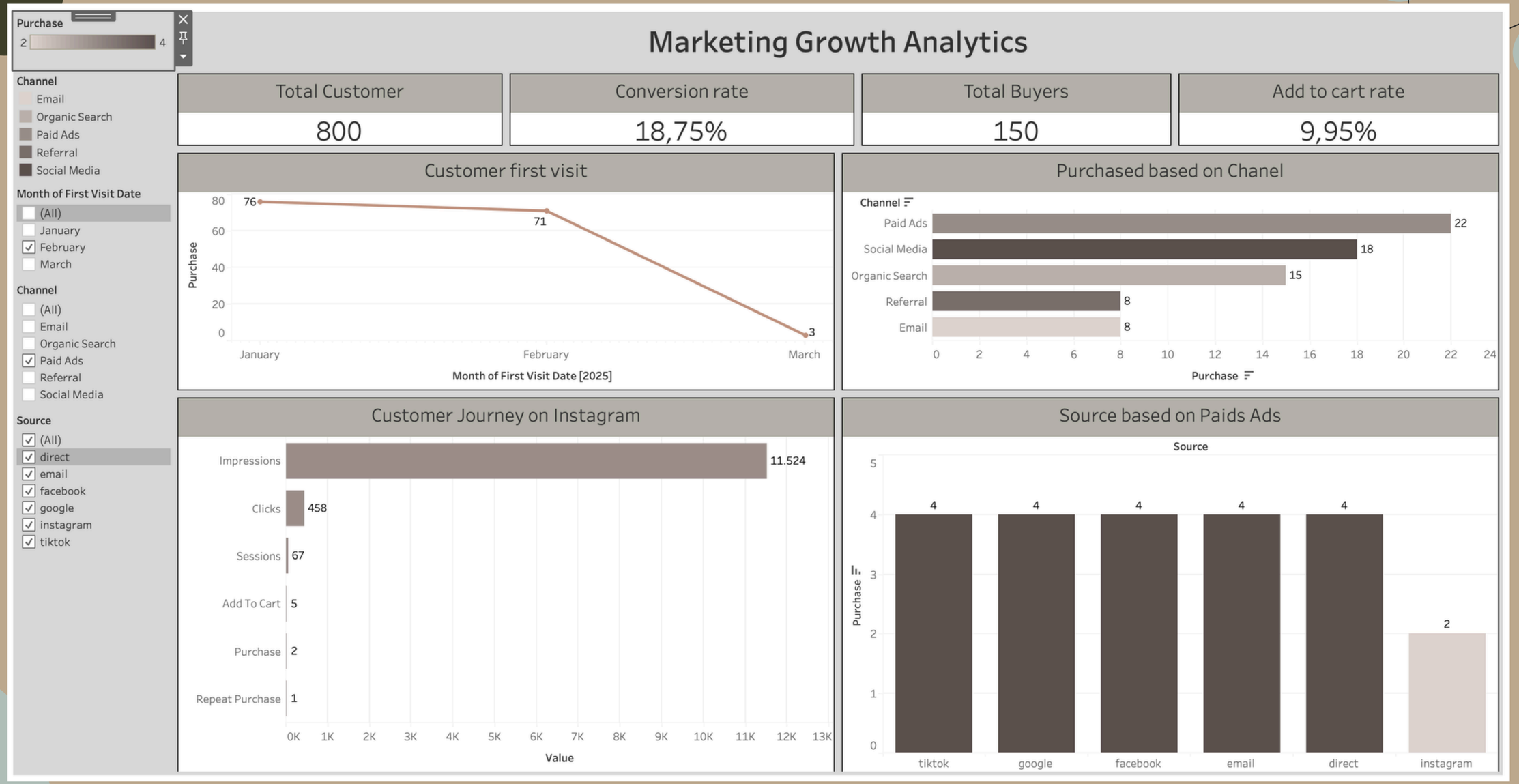
```
marketing_copy['first_visit_date'] = pd.to_datetime(marketing_copy['first_visit_date'])
```

```
marketing_copy.info()
```

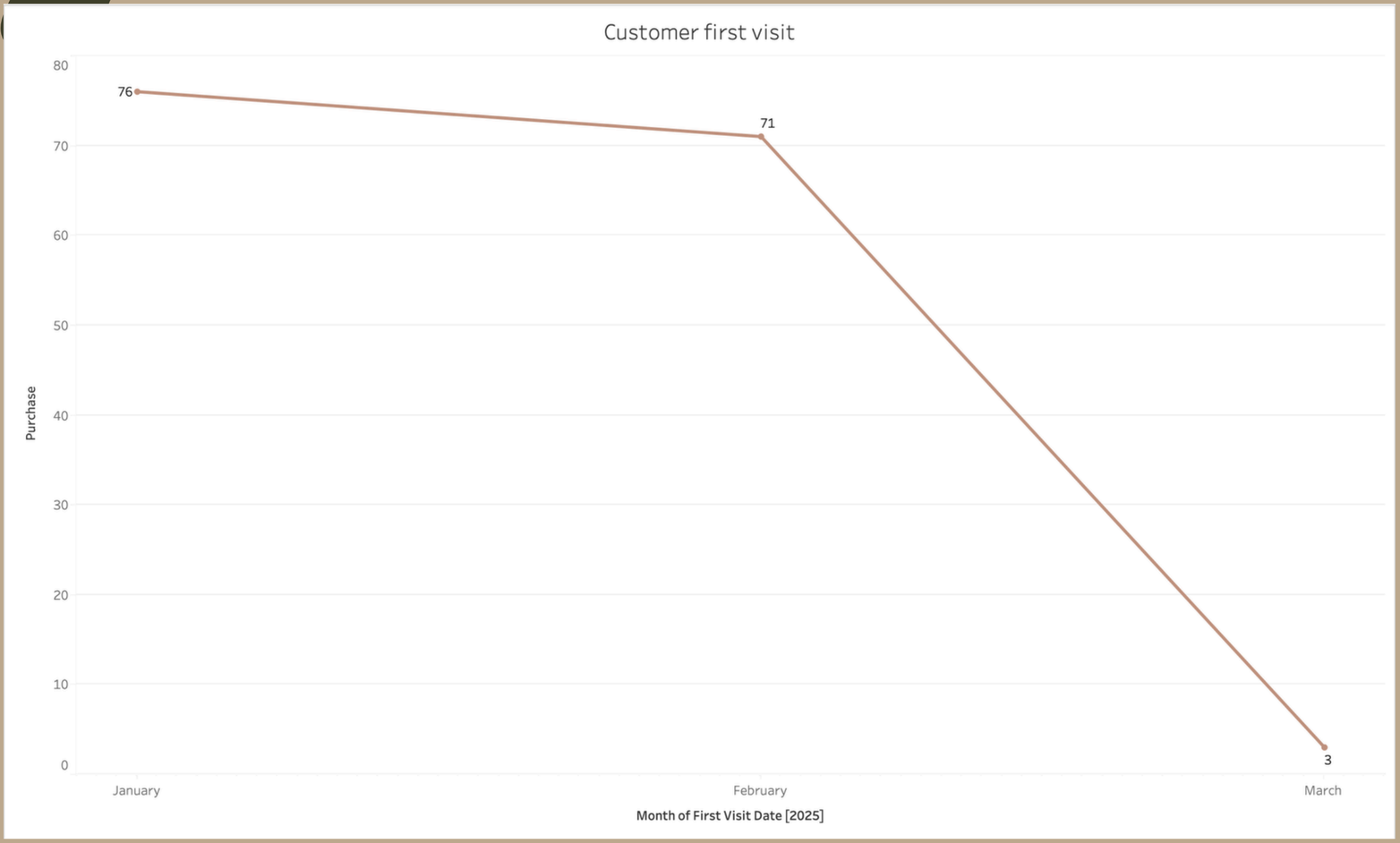
```
<class 'pandas.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 16 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   user_id           800 non-null    int64
 1   first_visit_date  800 non-null    datetime64[us]
 2   channel           800 non-null    str
 3   source            800 non-null    str
 4   device            800 non-null    str
 5   sessions          800 non-null    int64
 6   page_views        800 non-null    int64
 7   bounce            800 non-null    int64
 8   add_to_cart       800 non-null    int64
 9   purchase          800 non-null    int64
 10  revenue           800 non-null    int64
 11  ad_cost           800 non-null    int64
 12  clicks            800 non-null    int64
 13  impressions       800 non-null    int64
 14  repeat_purchase   800 non-null    int64
 15  retention_month   800 non-null    int64
dtypes: datetime64[us](1), int64(12), str(3)
memory usage: 117.9 KB
```

Penanganan untuk merubah tipe data dari string menjadi datetime

Data Visualization Output

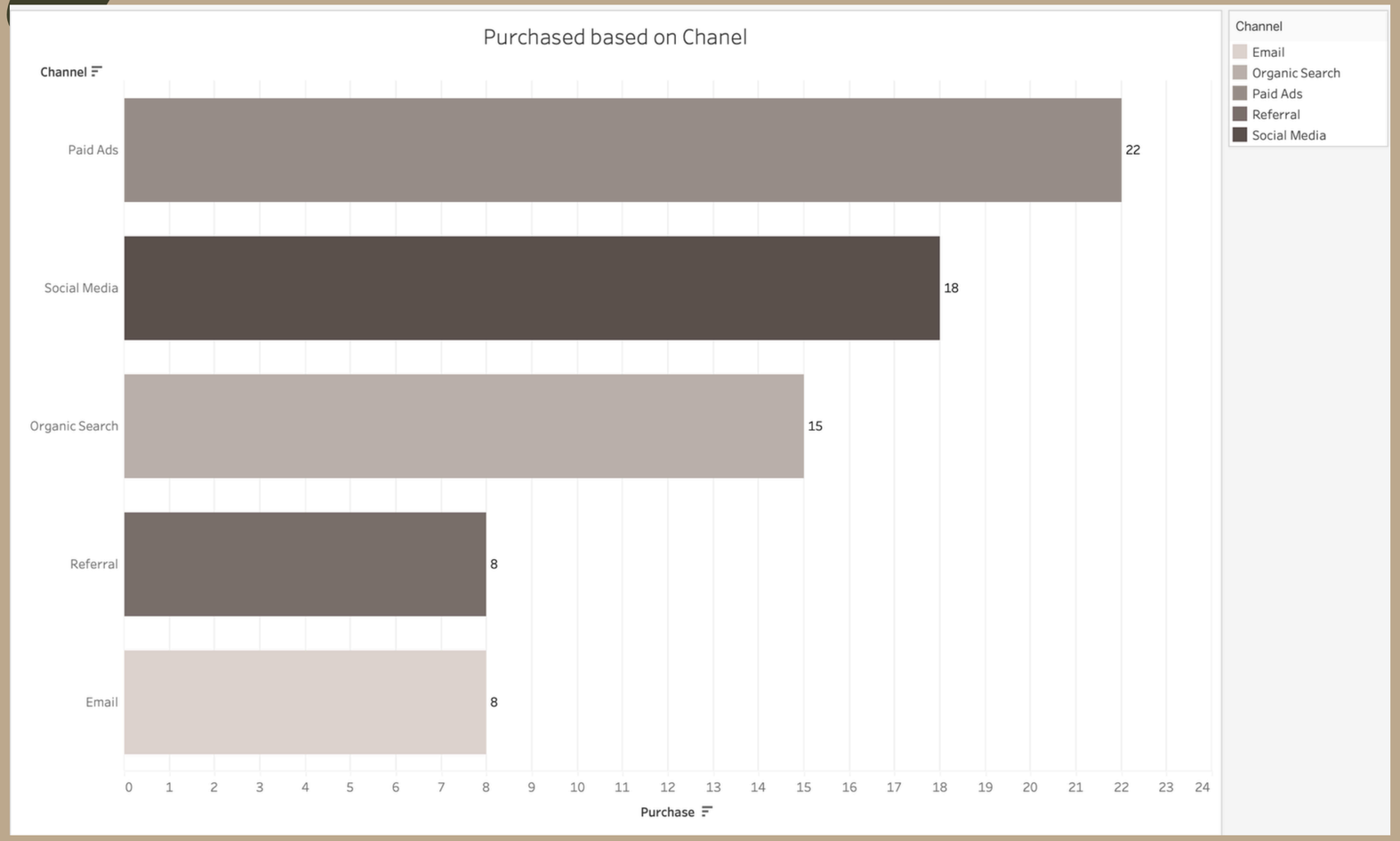


Data Analysis Output



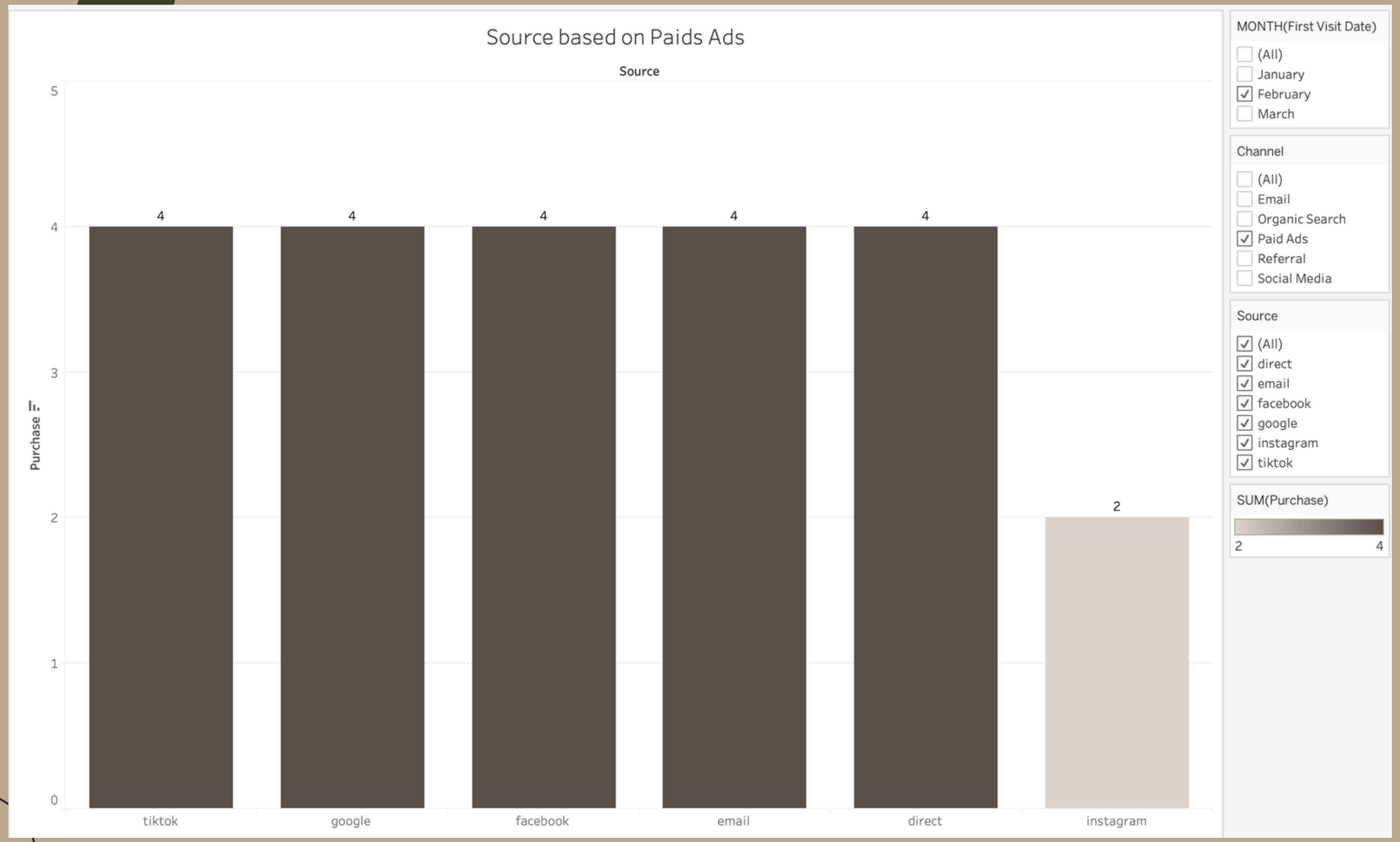
Terjadi penurunan untuk customer visit yang melakukan pembelian pada bulan Februari

Data Analysis Output



Paid Ads menghasilkan pembelian tertinggi, yaitu 22 pembelian, diikuti Social Media sebanyak 18 dan Organic Search sebanyak 15. Referral dan Email masing-masing menghasilkan 8 pembelian.

Data Analysis Output



Terjadi perbedaan khususnya source pada instagram berdasarkan paids ads untuk customer yang melakukan pembelian

Data Analysis Output

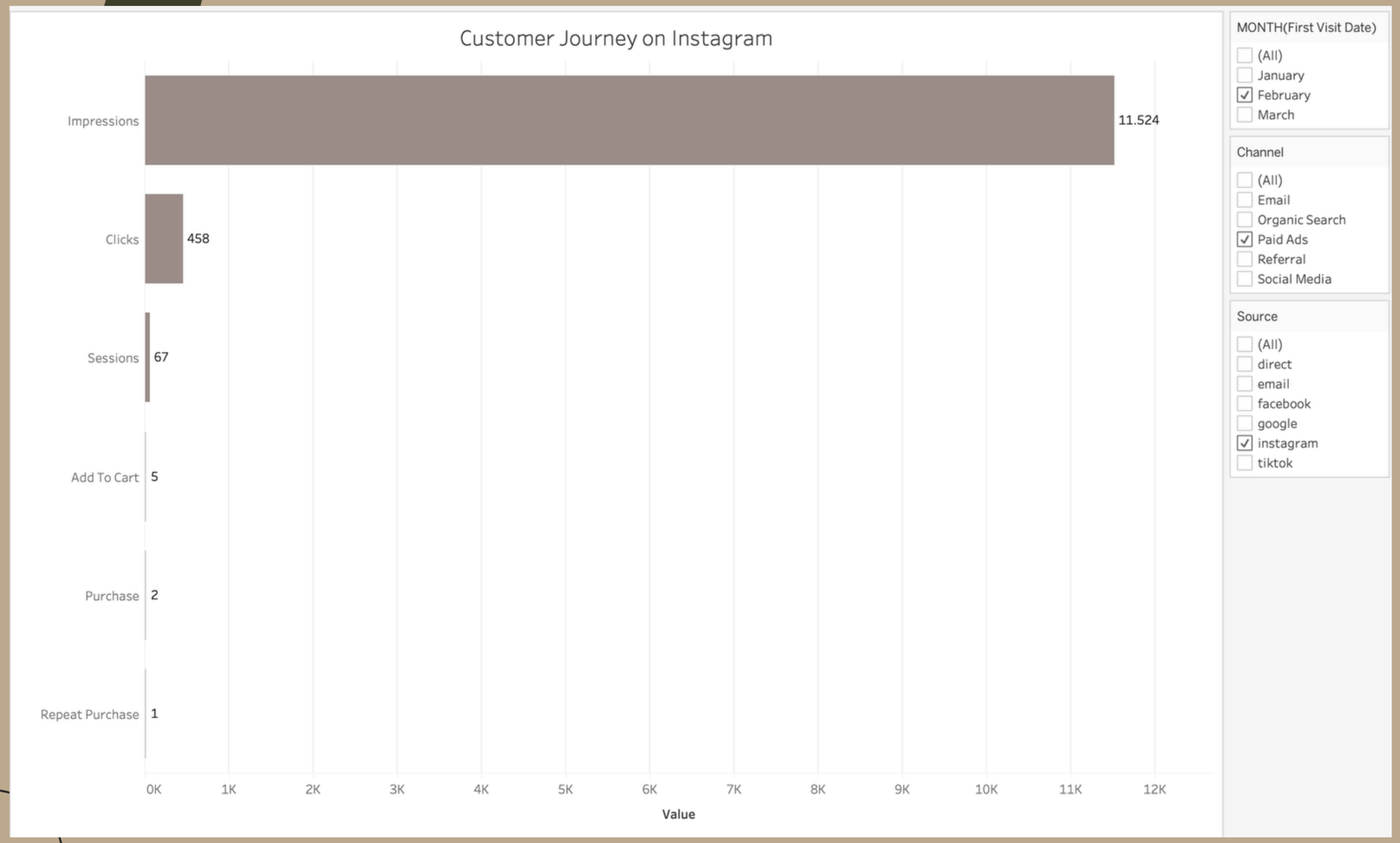


Chart menampilkan 11.524 impressions, 458 clicks, 67 sessions, 5 add to cart, 2 purchases, dan 1 repeat purchase pada customer journey Instagram. Penurunan jumlah terbesar secara signifikan terjadi dari Impressions ke Clicks, yaitu sebanyak 11.066. Sementara itu, dari 5 add to cart hanya 2 yang menjadi pembelian.

Marketing Strategy

Conclusion & Recommendations



Optimasi channel Paid Ads karena menghasilkan jumlah pembelian tertinggi dibandingkan channel lainnya, serta penargetan audiens yang lebih sesuai dengan karakteristik calon pelanggan

Memastikan bahwa landing page memiliki informasi yang jelas, waktu pemuatan yang cepat, serta pengalaman pengguna yang mudah dipahami.

Program loyalitas dan penawaran khusus dapat dikembangkan untuk mendorong pembelian berulang.

Thank You

Contact Person

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https://github.com/Alfi-Syahputra/Marketing-Growth-Analitycs

